

Sama Al-Oda

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Education

McMaster University

Expected May 2028

Bachelor of Software Engineering, Double Major with Biomedical Engineering – GPA: 3.96

Experience

Machine Learning Engineering Intern (CyberDome / R&D)

May 2026 – Aug 2026

Nokia | C, Python, eBPF, Kubernetes, OpenShift, Docker, Helm, Jenkins, Ansible

Ottawa, ON

- Productionized a **real-time behavioral ML security system**, debugging its C/eBPF telemetry layer and reducing anomaly alerts from **580+ alerts/s to approximately 1 alert/s** with zero reported kernel-event loss.
- Owned the system's **end-to-end MLOps lifecycle**, automating builds, air-gapped packaging, testing, and deployment of distributed ML services across **Kubernetes and OpenShift**.
- Built a **secure observability pipeline** connecting C/Python agents to enterprise SIEM systems, resolving networking, mTLS, and routing failures across **four deployment configurations**.

Machine Learning & Software Developer Intern

May 2025 – Apr 2026

Didar Lab / FendX Technologies | PyTorch, TensorFlow, Swift, AWS, PostgreSQL

Hamilton, ON

- Architected and deployed scalable cloud-based ML applications utilizing **PyTorch and TensorFlow**, engineering robust inference services through AWS-backed REST APIs to consistently achieve **sub-200ms response latency**.
- Fine-tuned advanced transformer-based and computer vision architectures using **Hugging Face**, leveraging large-scale multimodal datasets to significantly enhance the accuracy and reliability of predictive analytical models.
- Engineered robust cross-platform mobile applications for **iOS and Android** environments using **Swift and Gradle**, integrating **CreateML, PostgreSQL, Supabase, and Firebase** to support concurrent multi-user ML workflows.
- Designed and implemented automated **computer vision detection pipelines**, facilitating high-throughput data analysis and accelerating the transition of complex research prototypes into reliable enterprise-grade software solutions.

Projects & Hackathons

VALID: Multimodal Predictive AI (CUCAI Responsible AI Award) | BERT, XGBoost, FastAPI

- Won the **Responsible AI Award** at CUCAI 2026 for leading a team of 10 to develop a multimodal ML framework for advanced decision support, achieving **91% precision** and **88% macro-recall** on complex datasets of over **120,000 unstructured text records**.
- Architected a high-performance NLP pipeline using a **BERT-based transformer model** for dynamic semantic label extraction, deploying the ensemble model (**XGBoost, MLP, Random Forest**) via a **FastAPI** RESTful gateway.

Disaster Detection App (MEC 2nd Place) | Swift, iOS, REST APIs, Twilio

- Developed an iOS application leveraging on-device sensors and a disaster dataset API to detect hazardous events and trigger emergency alerts in real time with **<5s detection latency**.
- Integrated Twilio SMS services for instant notifications and implemented local data logging via caching to ensure offline reliability during critical events.

KareBear: Full-Stack Interactive App (ElleHacks 2nd Place) | React Native, GCP, TypeScript

- Engineered a cross-platform mobile application using **React Native and Expo**, implementing full-stack integrations with **Google Cloud APIs** to facilitate real-time, low-latency audio and visual processing, earning 2nd place out of 100+ teams.
- Architected hardware-to-cloud communication pipelines, securely routing physical sensor data to a **Gemini-powered conversational backend** to generate dynamic, interactive system responses.

Technical Skills

Machine Learning: PyTorch, TensorFlow, Keras, Hugging Face, BERT, XGBoost, Scikit-learn, OpenCV, CreateML, CoreML

Languages: Python, Swift, Java, JavaScript, TypeScript, SQL, C/C++, Bash, MATLAB

Data & Cloud: PostgreSQL, AWS (S3, EC2), Google Cloud (GCP), Supabase, Firebase

Software & Architecture: FastAPI, Flask, React.js, Node.js, Spring Boot, REST APIs, Microservices, gRPC, Pytest, JUnit

Infrastructure & Tools: Kubernetes, Docker, Podman, Helm, Ansible, Jenkins, GitLab CI/CD, Wazuh, Linux, Git, Postman